

Predicting U.S. Airline Arrival Delays Using Machine Learning

Business Problem

Flight delays affect airlines, airports, and passengers by disrupting schedules, increasing operating costs, contributing to missed connections, and reducing customer satisfaction. Historical flight data may help identify flights with an elevated risk of arriving late and support more proactive decision-making.

This project examines whether information available before departure can predict whether a U.S. domestic flight will arrive at least 15 minutes late. Predictors include scheduled departure and arrival times, month, day of week, airline, origin and destination airports, scheduled flight duration, and distance. Logistic Regression and Random Forest models are evaluated to determine whether these characteristics can distinguish delayed from non-delayed flights.

The goal is not to create a production-ready system, but to evaluate the usefulness of routinely available flight information for predicting significant arrival delays.

Background and History

The U.S. commercial aviation system handles more than 16 million flights annually and approximately 44,000 flights on an average day (Federal Aviation Administration, n.d.). Because airlines, airports, aircraft, crews, and air traffic operations form an interconnected network, disruptions can affect subsequent flights and contribute to delays.

Flight delays may result from airline operations, weather, airport congestion, air traffic management, security events, and late-arriving aircraft. The Bureau of Transportation Statistics (BTS) maintains the Reporting Carrier On-Time Performance database, which contains individual domestic flight records including scheduled and actual flight times, airports, cancellations, diversions, delay causes, air time, and distance (Bureau of Transportation Statistics, n.d.). The U.S. Department of Transportation also publishes consumer reports using flight-performance information collected by BTS (U.S. Department of Transportation, 2026a).

Data Explanation and Preparation

The study used the BTS Reporting Carrier On-Time Performance database. Twelve monthly files covering January through December 2025 were downloaded and combined using Python and pandas. The resulting dataset contained 7,001,619 flight records and 34 selected variables.

The initial audit found no duplicate rows. There were 102,876 cancelled flights (1.47%), 19,258 diverted flights (0.28%), and 122,135 records without a valid ARR_DEL15 outcome. Because the research question focuses on completed flights with known arrival outcomes, cancelled, diverted, and invalid-target records were excluded.

The final analytical dataset contained 6,879,484 completed flights. Of these, 5,344,846 (77.69%) arrived on time or less than 15 minutes late, while 1,534,638 (22.31%) arrived at least 15 minutes late.

The target variable was ARR_DEL15. Predictors were restricted to information available before departure: month, day of week, airline, origin, destination, scheduled departure and arrival times, scheduled elapsed time, and distance. Actual delay measures and delay-cause variables were excluded from predictive modeling to prevent data leakage.

Methods

Exploratory data analysis examined arrival-delay rates by month, carrier, origin airport, scheduled departure period, day of week, and flight distance. Delay rates varied substantially throughout 2025. July had the highest monthly rate at 28.89%, followed by June at 28.26%, while September had the lowest rate at 16.63%.

Scheduled departure time produced one of the strongest patterns. Early-morning flights had an 11.00% delay rate, compared with 16.69% in the morning, 25.38% in the afternoon, 31.82% in the evening, and 27.44% late at night.

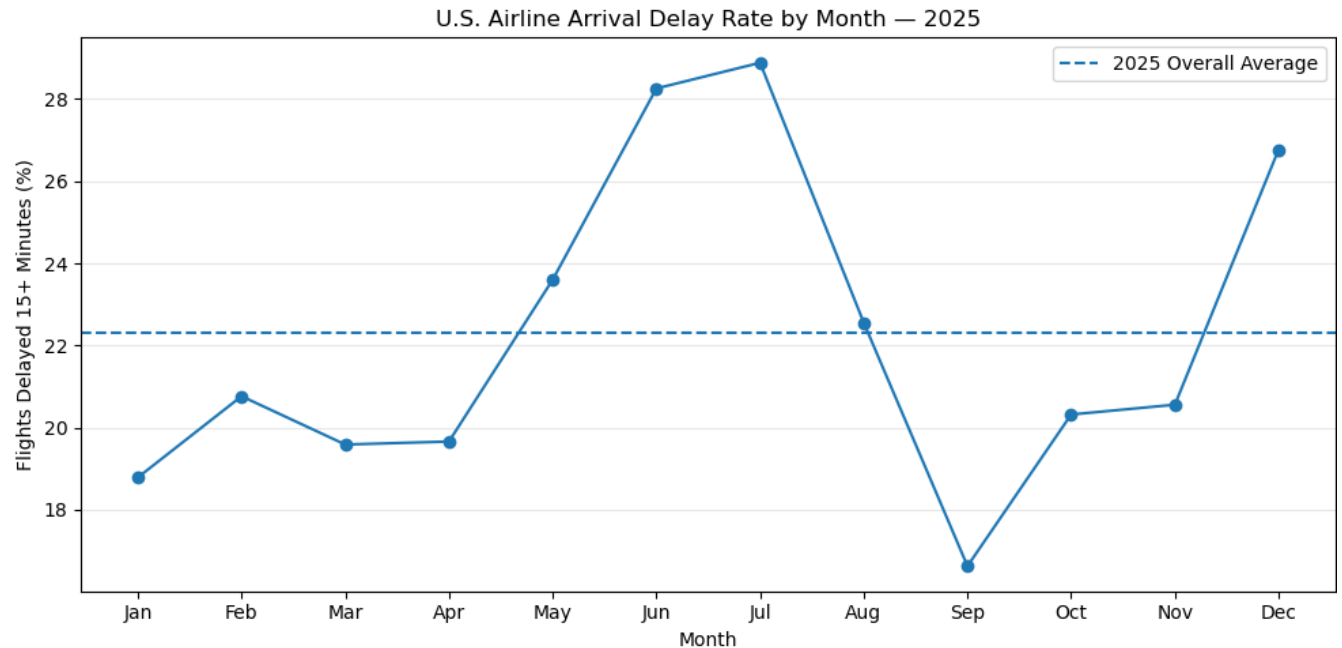


Figure 1. U.S. Airline Arrival Delay Rate by Month — 2025.

Day of week also showed differences, ranging from 18.39% on Tuesday to 26.36% on Sunday. Flight distance showed a weaker relationship, with delay rates ranging from 20.69% to 23.34% across the distance categories examined.

For modeling, a stratified sample of 1,000,000 flights was selected and divided into 800,000 training and 200,000 test observations, preserving the 22.31% delay rate. A majority-class baseline, Logistic Regression, and Random Forest were compared. Categorical predictors were one-hot encoded, and balanced class weighting was applied to the machine-learning models. Performance was evaluated using accuracy, precision, recall, F1 score, and ROC-AUC.

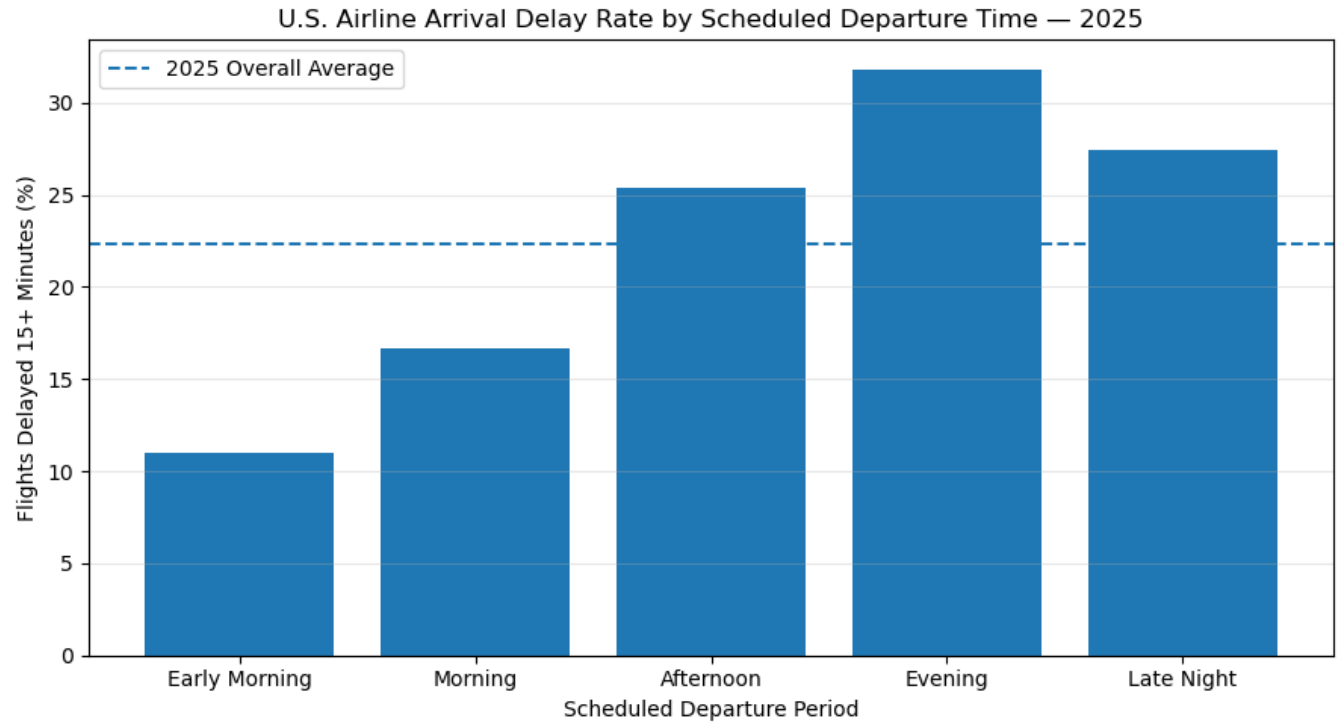


Figure 2. U.S. Airline Arrival Delay Rate by Scheduled Departure Time — 2025.

Analysis and Results

The baseline achieved 77.69% accuracy by predicting every flight as not delayed. However, it identified no delayed flights, resulting in zero recall and F1 for the delayed class. This demonstrates why accuracy alone is misleading for an imbalanced target.

Logistic Regression achieved 60.56% accuracy, 30.99% precision, 62.58% recall, a 0.415 F1 score, and 0.651 ROC-AUC. Random Forest achieved 60.22% accuracy, 31.08% precision, 64.37% recall, a 0.419 F1 score, and 0.668 ROC-AUC.

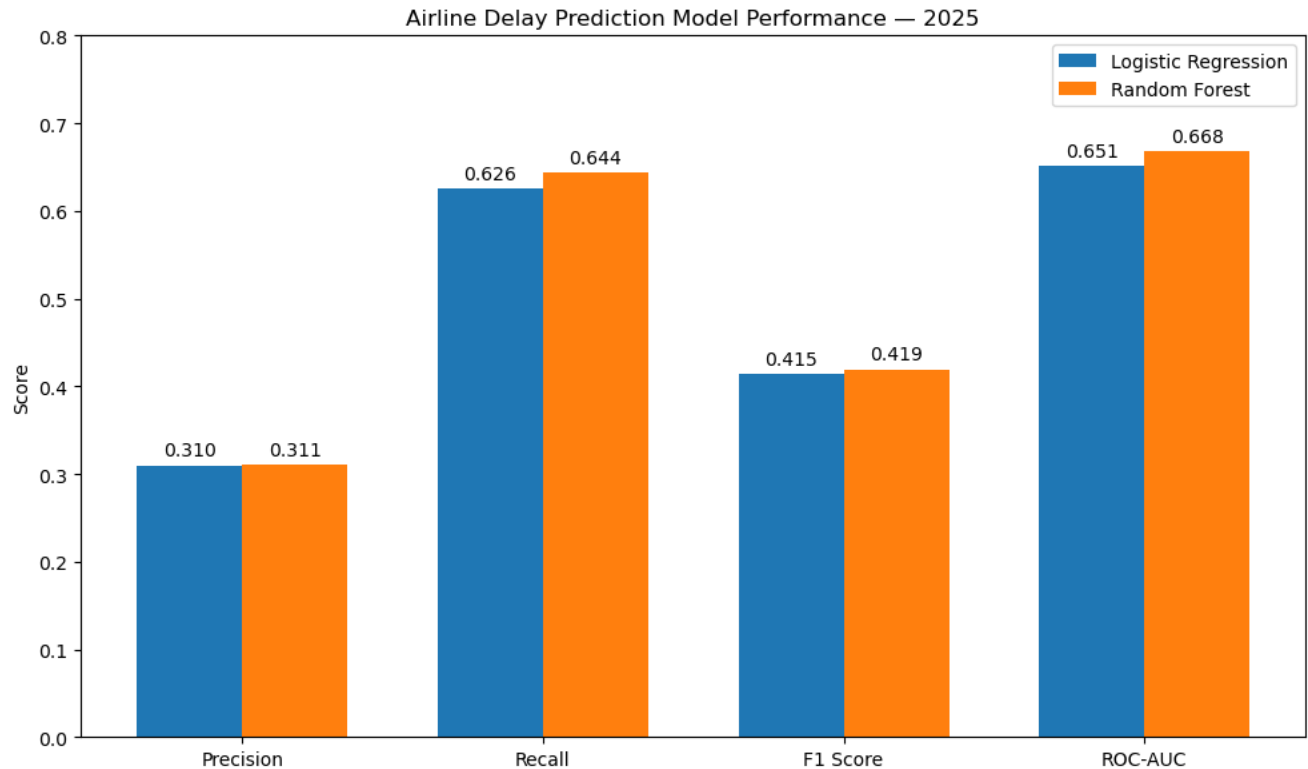


Figure 3. Airline Delay Prediction Model Performance — 2025.

Random Forest was the strongest model based on recall, F1, and ROC-AUC. Its recall indicates that approximately 64% of actual delayed flights were identified. However, precision was only 31.08%, meaning many flights predicted as delayed were false positives. The model therefore demonstrates useful but limited predictive capability.

Random Forest feature importance showed that scheduled arrival and departure times were the most influential predictors, with importance values of approximately 0.323 and 0.320. Month and day of week followed, while scheduled elapsed time, distance, airline, and airport variables provided additional information.

TOP 20 RANDOM FOREST FEATURES		
	Feature	Importance
720	num_CRS_ARR_TIME	0.323309
719	num_CRS_DEP_TIME	0.319589
717	num_MONTH	0.070901
718	num_DAY_OF_WEEK	0.034149
721	num_CRS_ELAPSED_TIME	0.025026
722	num_DISTANCE	0.014694
104	cat_ORIGIN_DFW	0.013536
0	cat_OP_UNIQUE_CARRIER_AA	0.013467
3	cat_OP_UNIQUE_CARRIER_DL	0.008358
12	cat_OP_UNIQUE_CARRIER_WN	0.008033
4	cat_OP_UNIQUE_CARRIER_F9	0.007722
10	cat_OP_UNIQUE_CARRIER_OO	0.006754
9	cat_OP_UNIQUE_CARRIER_OH	0.006172
676	cat_DEST_SLC	0.006103
255	cat_ORIGIN_ORD	0.005971
324	cat_ORIGIN_SLC	0.005263
200	cat_ORIGIN_LAX	0.005042
451	cat_DEST_DCA	0.004464
165	cat_ORIGIN_HNL	0.004411
607	cat_DEST_ORD	0.003956

Figure 4. Top Random Forest Predictors of Airline Arrival Delays — 2025.

These results are consistent with the EDA, particularly the substantial variation by scheduled departure period. Feature importance represents contribution to prediction and should not be interpreted as evidence of causation.

Conclusion

Pre-departure flight characteristics contain useful but limited information for predicting significant arrival delays. Among the 6,879,484 completed flights, 22.31% arrived at least 15 minutes late. Delay rates varied notably by month, day of week, carrier, airport, and especially scheduled departure time.

Random Forest produced the strongest predictive results, with a 0.668 ROC-AUC and 64.37% recall. However, its low precision indicates a substantial false-positive rate. The model is therefore useful for demonstrating predictive patterns but does not provide sufficiently strong predictive performance to serve as a stand-alone operational forecasting system.

Assumptions

The analysis assumes that BTS records are sufficiently accurate and consistently reported for aggregate analysis and that ARR_DEL15 appropriately identifies significant arrival delays. It also assumes that historical relationships observed in 2025 contain information relevant to delay risk, although these relationships may change over time.

The predictive analysis further assumes that the selected variables represent information reasonably available before departure.

Limitations

The analysis covers only one calendar year, so the findings may not fully generalize to future operating conditions. Important real-time predictors—including weather, airport congestion, air traffic restrictions, and incoming-aircraft status—were not included.

Models were developed using a stratified one-million-flight sample and a random train/test split. A future system should use temporal validation by training on earlier periods and evaluating performance on later, unseen flights. Finally, the analysis identifies predictive associations rather than causal relationships.

Challenges

The primary technical challenge was processing more than seven million records across 12 monthly files. Efficient variable selection and sampling were necessary to manage computational requirements.

Class imbalance was another challenge because only 22.31% of completed flights were significantly delayed. This made accuracy misleading and required evaluation using precision, recall, F1, and ROC-AUC. Preventing data leakage also required separating variables useful for historical explanation from those appropriate for pre-departure prediction.

Future Uses and Additional Applications

Future research could integrate weather forecasts, airport congestion, air traffic restrictions, and aircraft-rotation information. Adding multiple years of BTS data would also allow models to be trained on historical periods and tested on future periods.

Additional applications could include predicting arrival-delay duration in minutes, severe delays, cancellations, diversions, or departure delays rather than using only a binary arrival-delay outcome.

Recommendations

Schedule-related characteristics should receive particular attention in future delay-prediction systems. Scheduled departure and arrival times were the most influential Random Forest predictors, while EDA showed substantially lower delay rates for early-morning flights.

However, the current model should be used only as a decision-support tool. Its low precision means predictions should supplement operational expertise and real-time information rather than independently determine operational actions.

Implementation Plan

A practical system could combine scheduled flight information with historical BTS records to generate a pre-departure probability of significant arrival delay. Future versions should integrate real-time weather, airport congestion, air traffic, and aircraft-status data.

Before deployment, the model should undergo temporal validation on future flights. Prediction thresholds could then be adjusted according to the relative cost of false positives and missed delays. Performance should be monitored continuously, with periodic retraining as operating conditions change.

Ethical Assessment

Predictive airline analytics require careful communication because incorrect predictions could influence passenger behavior or operational decisions. Model outputs should therefore be presented as probabilities rather than guarantees, with limitations and error rates clearly disclosed.

Airline and airport comparisons also require context. Higher delay rates do not necessarily indicate poorer management because carriers and airports operate under different weather, geographic, traffic, scheduling, and network conditions. Predictive associations should not be interpreted as causal judgments about organizational quality.

Finally, automated predictions should supplement rather than replace qualified human judgment, particularly for safety-related or other high-impact aviation decisions.

10 Potential Audience Questions and Answers

1. Why did the baseline model have higher accuracy than the machine-learning models?

The dataset was imbalanced, with 77.69% of completed flights classified as not delayed. The baseline simply predicted every flight as not delayed, producing 77.69% accuracy but identifying none of the delayed flights. Logistic Regression and Random Forest had lower overall accuracy but detected approximately 63% and 64% of delayed flights, respectively. For this reason, recall, F1 score, and ROC-AUC provide more meaningful measures of predictive performance than accuracy alone.

2. Why was a 15-minute threshold used to define a delayed flight?

The target variable, ARR_DEL15, follows the Bureau of Transportation Statistics definition of an arrival delay of 15 minutes or more. Using this established measure provides a consistent and interpretable definition of a significant delay.

3. Why were cancelled and diverted flights excluded?

The research question focuses specifically on predicting whether a completed flight will arrive at least 15 minutes late. Cancelled and diverted flights represent different operational outcomes and do not have a comparable arrival-delay classification. They would be better examined using separate predictive models.

4. Why were only one million flights used for model development when nearly 6.9 million completed flights were available?

A stratified sample of one million flights reduced computational requirements while retaining a very large modeling dataset. Stratification preserved the original 22.31% delayed-flight proportion in both the training and test sets.

5. Why were Logistic Regression and Random Forest selected?

Logistic Regression provides a widely used and interpretable classification benchmark, while Random Forest can capture nonlinear relationships and interactions among predictors. Comparing the two allowed the project to evaluate whether a more flexible ensemble method provided meaningful improvement over a simpler statistical model.

6. Why was Random Forest considered the stronger model even though its accuracy was slightly lower than Logistic Regression?

Random Forest produced higher recall, F1 score, and ROC-AUC. It identified 64.37% of delayed flights compared with 62.58% for Logistic Regression and achieved the highest ROC-AUC at 0.668. Because detecting delayed flights was the primary modeling objective, these measures were more informative than the small difference in overall accuracy.

7. Which variables were most important for predicting delays?

Scheduled arrival time and scheduled departure time were the two most influential Random Forest predictors. Month and day of week were also important. These findings were consistent with the exploratory analysis, which showed substantial differences in delay rates across departure periods and seasons.

8. What additional data could improve the model?

Real-time weather, airport congestion, air traffic restrictions, aircraft rotation information, and incoming flight status would likely provide valuable additional information. The current model intentionally uses scheduled and historical flight characteristics that are available before departure.

9. Could this model be used by an airline in its current form?

Not as a stand-alone operational system. Its 64.37% recall demonstrates useful predictive information, but its 31.08% precision means it produces many false-positive delay predictions. A production system would require additional predictors, temporal validation, probability calibration, threshold optimization, and continuous performance monitoring.

10. What are the primary ethical concerns associated with this type of model?

Predictions could influence passengers or operational decisions even when they are incorrect. Results should therefore be communicated as probabilities rather than guarantees. Comparisons among airlines and airports also require context because weather, geography, traffic volume, and network structure affect delay rates. Predictive associations should not be interpreted as evidence that a particular airline or airport causes delays.

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